

Linear Programming 1

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Introductory Example

A firm manufactures two products (only) at their Beeham factory. Both products require the use of the GG74 machine.

Product X requires 3 hours per unit at this machine

Product Y requires 2 hours per unit at this machine

In any week there are 144 hours of time available

Each unit of Product X gives a Contribution to Profit of £5, and

Each unit of Product Y gives a Contribution to Profit of £7.

Linear Programming Model

Constraints

Production Constraint

- Product X requires 3 hours per unit at this machine
- Product Y requires 2 hours per unit at this machine
- In any week there are 144 hours of time available

$$3x + 2y \leq 144$$

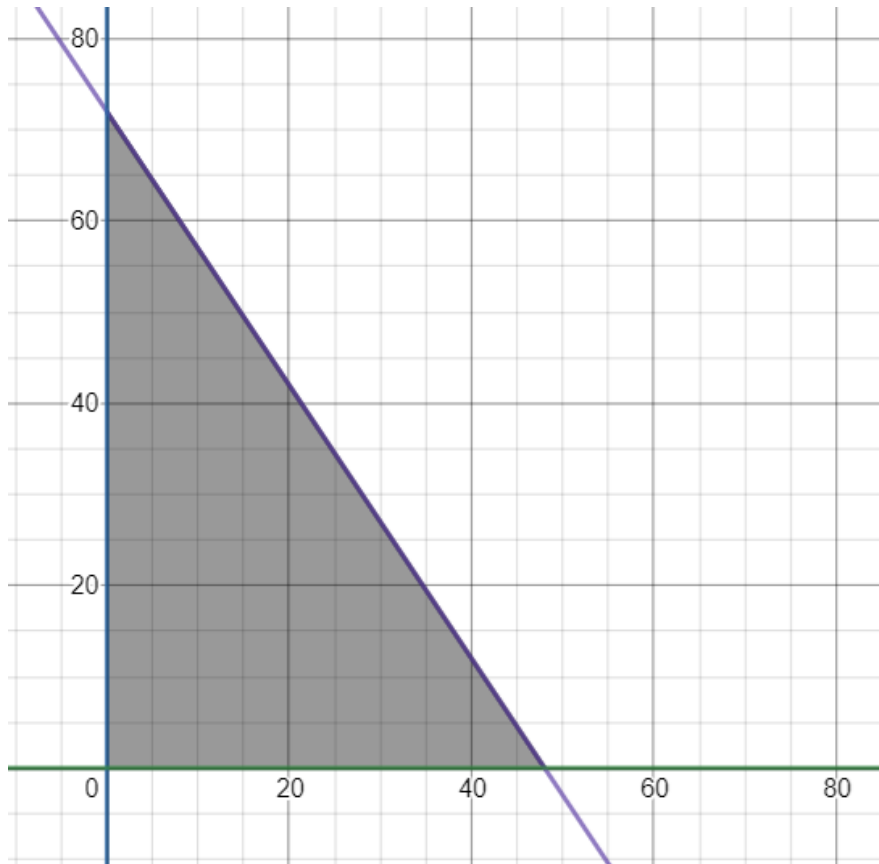
Non-negativity

- It is not possible to manufacture a negative amount!

$$\begin{aligned} x &\geq 0 \\ y &\geq 0 \end{aligned}$$

Linear Programming Model

Feasible Region



Any point within the feasible region is a solution to the system of inequalities.

$$3x + 2y \leq 144 \quad (\text{Constraints})$$

$$x \geq 0$$

$$y \geq 0$$

Linear Programming Model

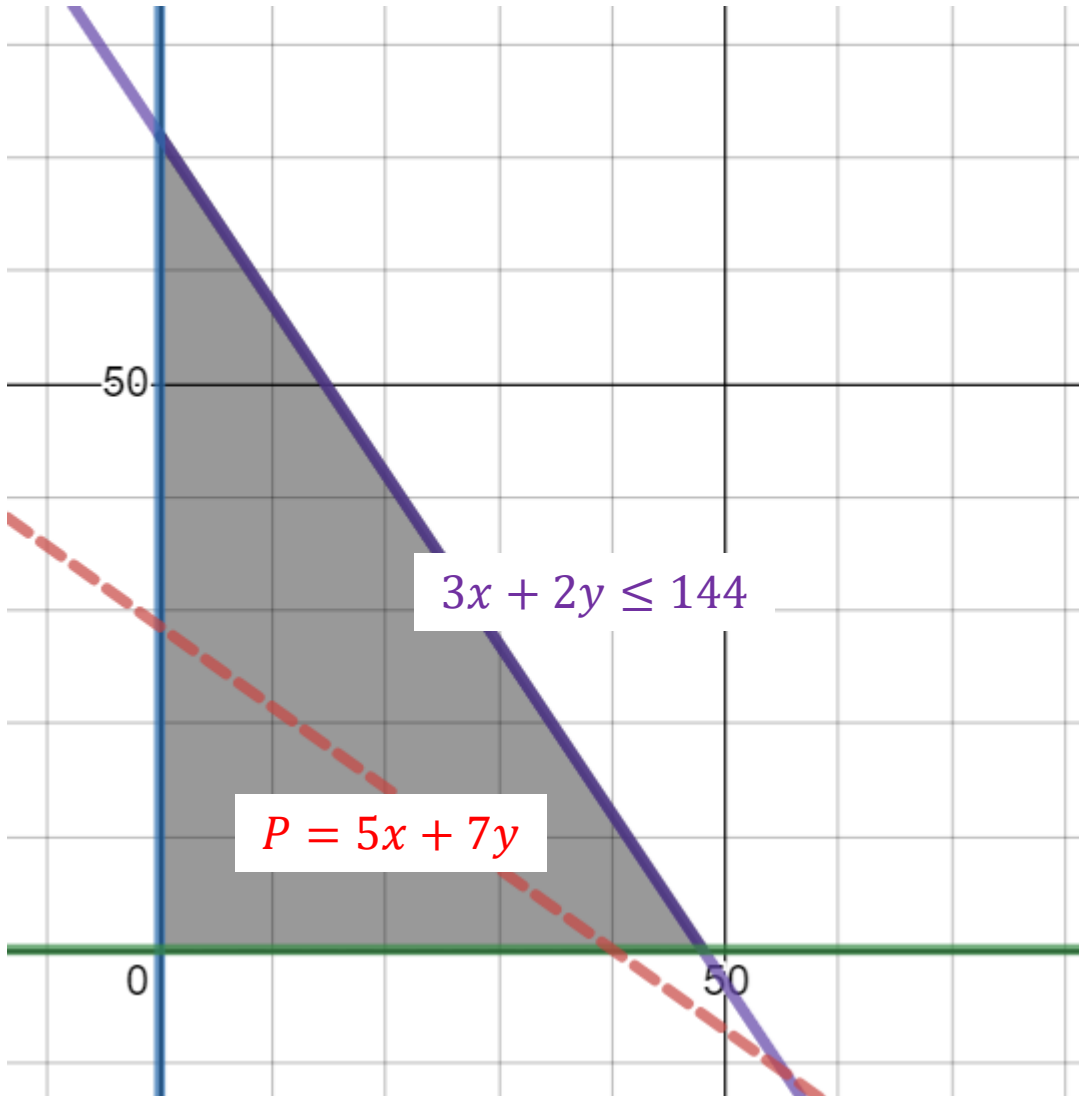
Add an Objective

Maximise the profit

Each unit of Product X gives a Contribution to Profit of £5, and
Each unit of Product Y gives a Contribution to Profit of £7.

$$P = 5x + 7y$$

Linear Programming Model



Graphically $P = 5x + 7y$ is a straight line.

e.g. a Profit of $P = 200$ is represented by

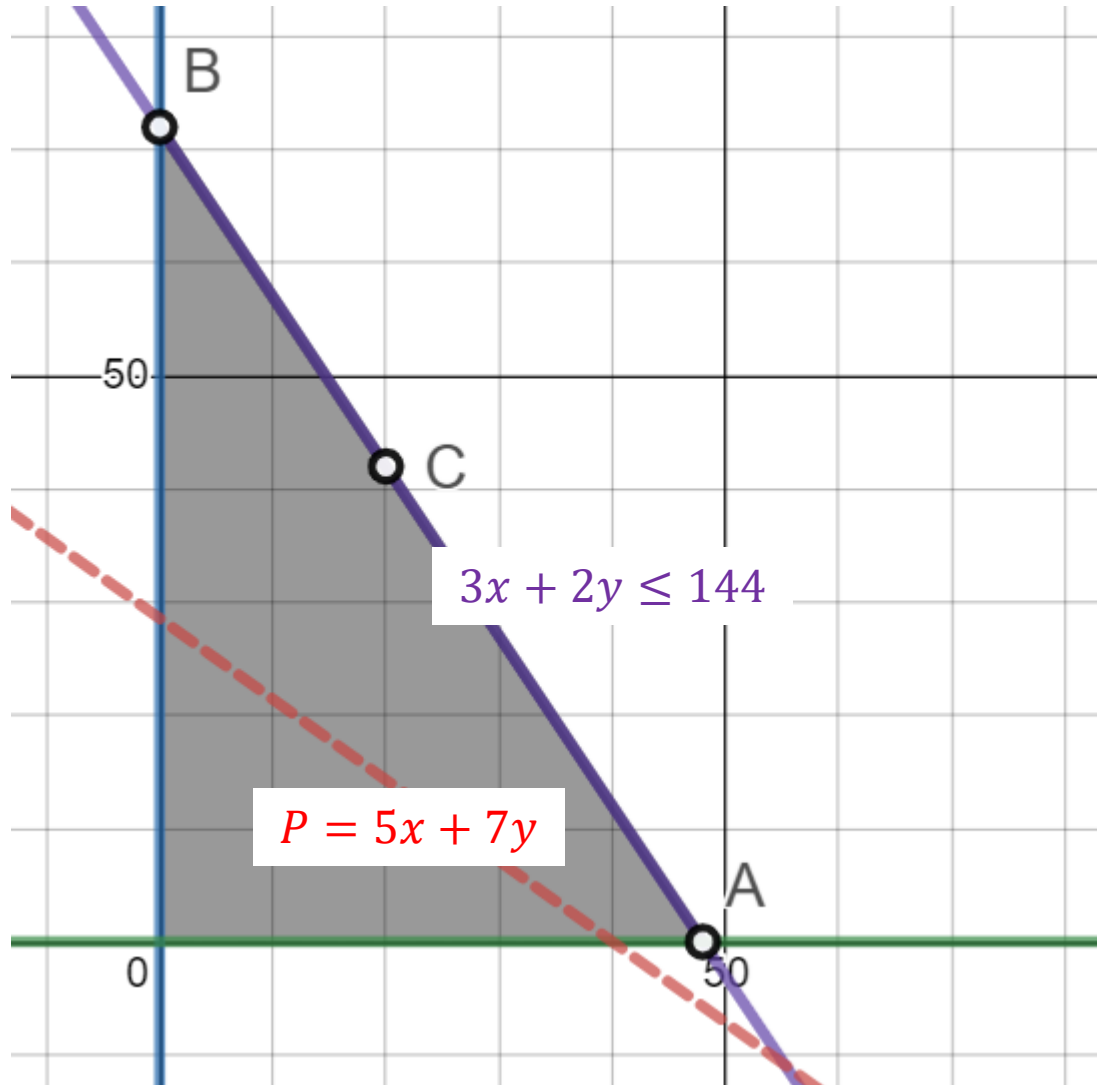
$$5x + 7y = 200$$

Choose a point (x, y) in the feasible region so as to maximise P

This is akin to moving the **red** profit line away from 0 as far as possible.

[Interactive demo](#)

Linear Programming Model



At A: $x = 48, y = 0, P = 240$

At B: $x = 0, y = 72, P = 504$

At C: $x = 20, y = 42, P = 394$

Because the functions are Linear the Profit rises along AB from A to B.

Algebraically:

$$y = \frac{144 - 3x}{2}$$

$$\begin{aligned} P &= 5x + 7\left(\frac{144 - 3x}{2}\right) \\ &= -5.5x + 504 \end{aligned}$$

We are allowed values of $0 \leq x \leq 48$, hence the maximum P is at $x = 0$. Point B!

Linear Programming Model

Our linear programming model can be written as follows

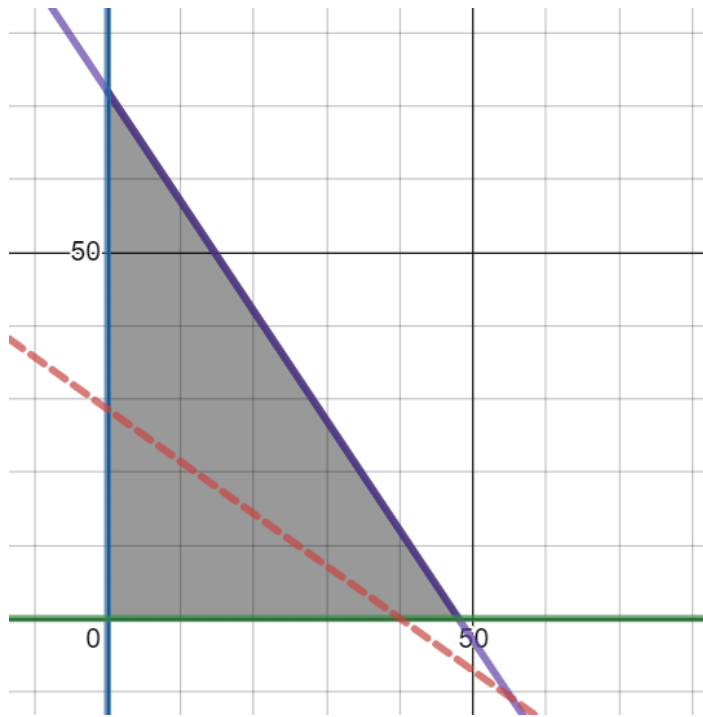
maximise $P = 5x + 7y$ (Objective)

subject to:

$3x + 2y \leq 144$ (Constraints)

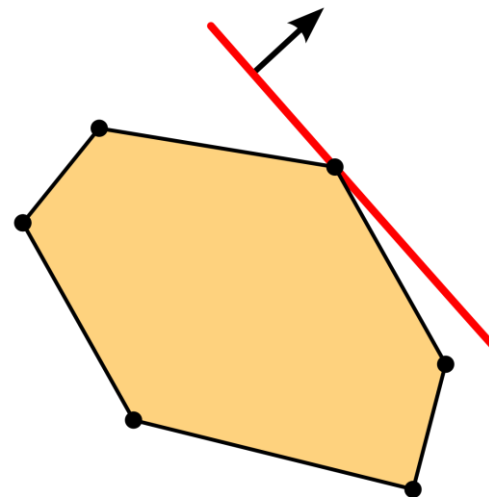
$x \geq 0$

$y \geq 0$



Observations

- We had 1 constraint and 2 variables.
 - The optimal solution had one variable set to 0...
- The solution was at a vertex (corner of the feasible region)
- Intuitively we are moving the objective function (linear equation) over the feasible region until we exit.



Is it a suitable solution?

QUESTION: Is this solution acceptable/workable for this Factory?

If not revisit the formulation.

For example, Should the number of x produced be related to the number of y produced?

(Chairs x and Tables y) – Perhaps through experience the manager knows that tables will not sell on their own.

Is it a suitable solution?

Introduce the constraint

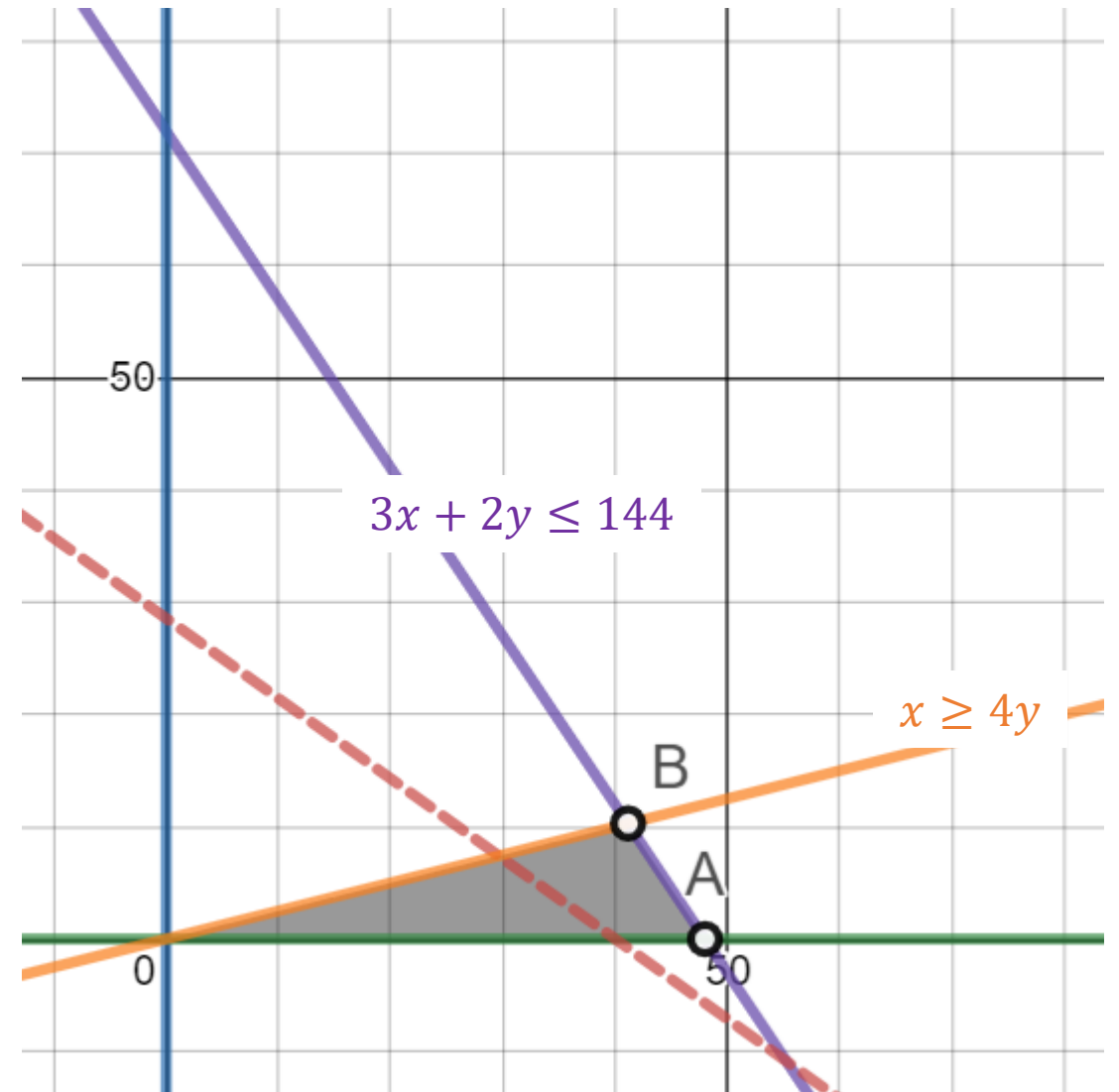
$$x \geq 4y$$

i.e. make enough tables to go with chairs

Our profit P will now be maximised at B.

$$(x, y) = (41.14, 10.29)$$
$$P = £277.72$$

Note: We now have 2 binding constraints and the 2 variables are both non-zero (i.e. used)



Linear Programming Model

Our linear programming model can be written as follows

maximise

$$P = 5x + 7y$$

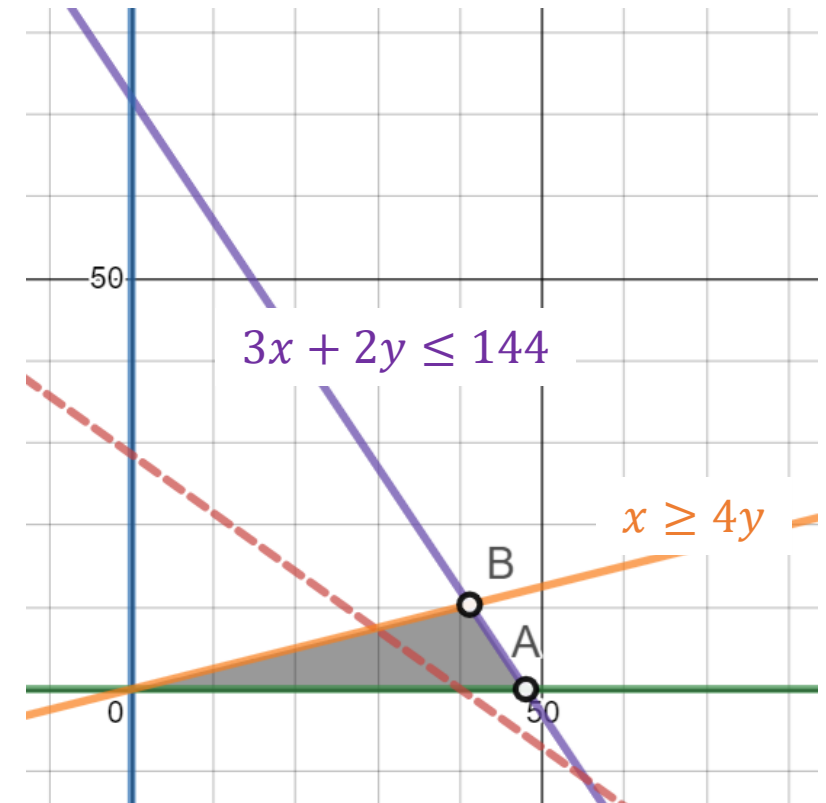
subject to:

$$3x + 2y \leq 144 \quad (\text{C1})$$

$$x \geq 4y \quad (\text{C2})$$

$$x \geq 0$$

$$y \geq 0$$



Binding Constraints

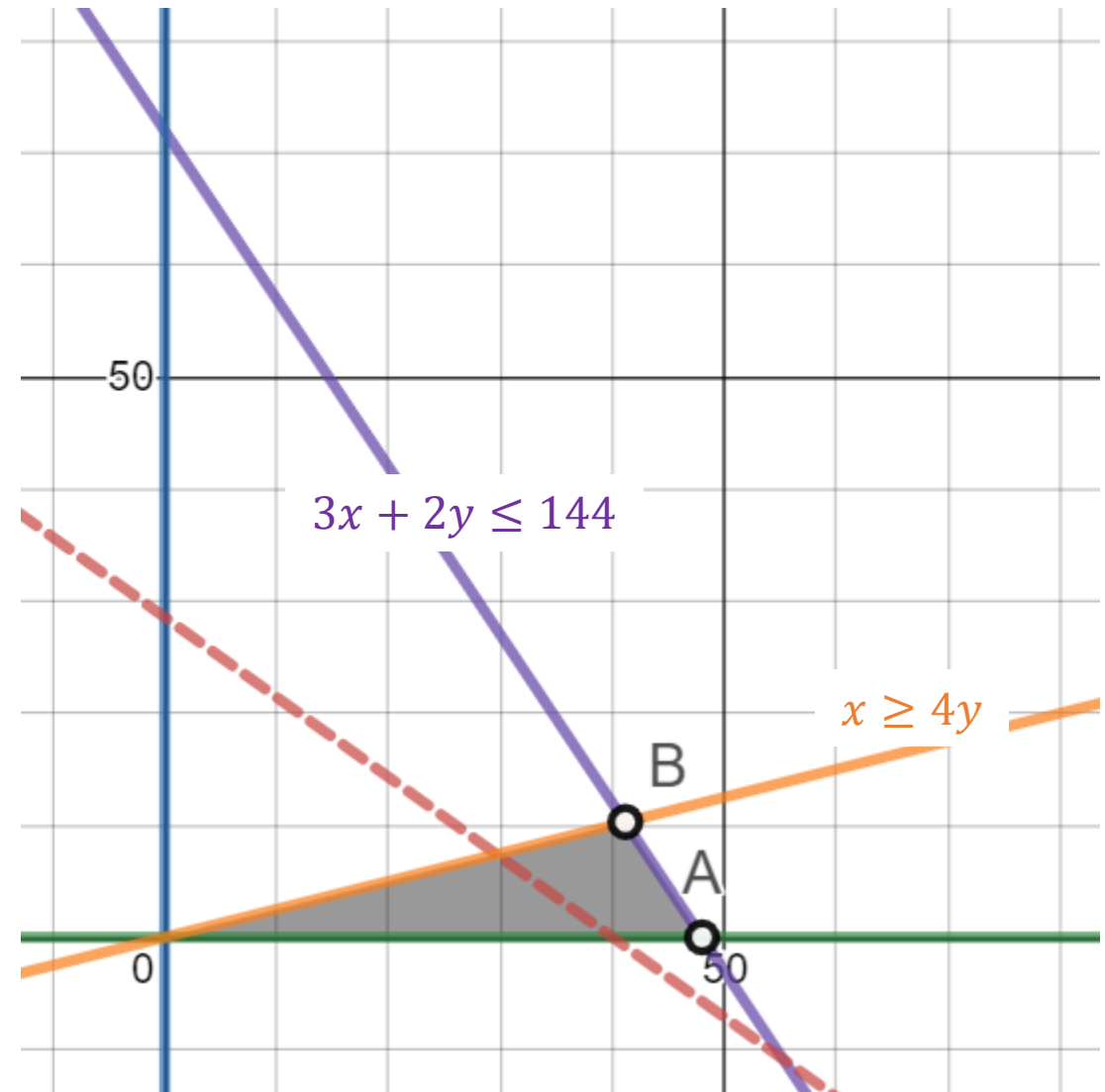
At **A**, $x = 48, y = 0$

$$3x + 2y = 144$$

Thus constraint **(C1)** is binding

$$48 \geq 4 \cdot 0$$

And constraint **(C2)** is not binding



Binding Constraints

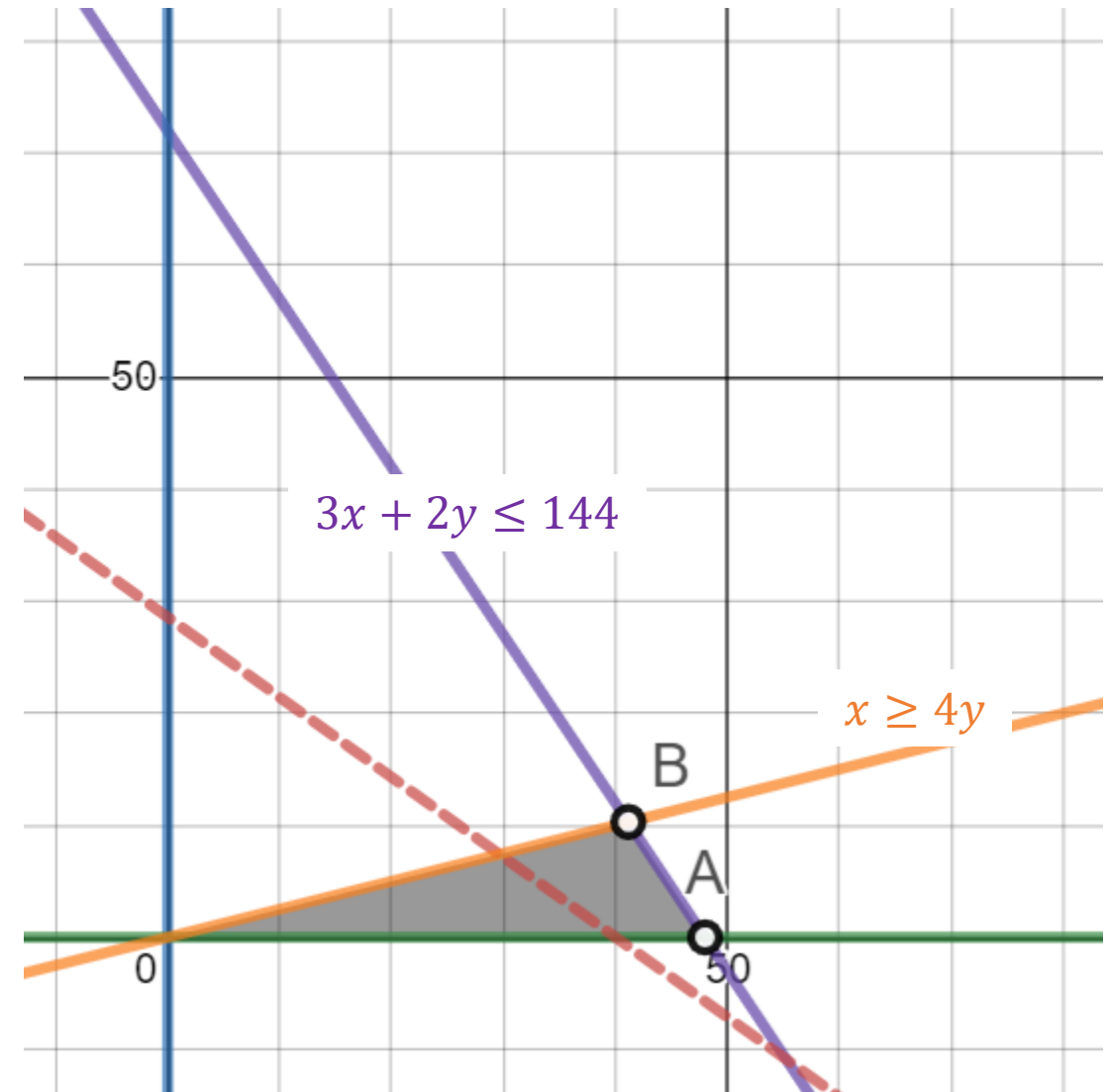
At **B**, $x = 41.14$, $y = 10.29$

$$3x + 2y = 144$$

Thus constraint **(C1)** is binding

$$41.14 = 4 \cdot 10.29$$

And constraint **(C2)** is also binding



Infeasible

What about if the manager demands making 60 of product x a week?

$$x \geq 60$$

We can't simultaneously satisfy

$$3x + 2y \leq 144$$
$$x \geq 60$$

$$\text{i.e. } 3 \cdot 60 = 180 > 144$$

This is known as an infeasible problem, there is no feasible region!

Constraints are inconsistent



Unbounded

What about if we remove both binding constraints

$$3x + 2y \leq 144$$
$$x \geq 4y$$

Our profit line $P = 5x + 7y$ is now free to move to an unlimited profit (grey feasible region).

This is known as an unbounded problem, there is no restriction on the objective



Infinite Solutions

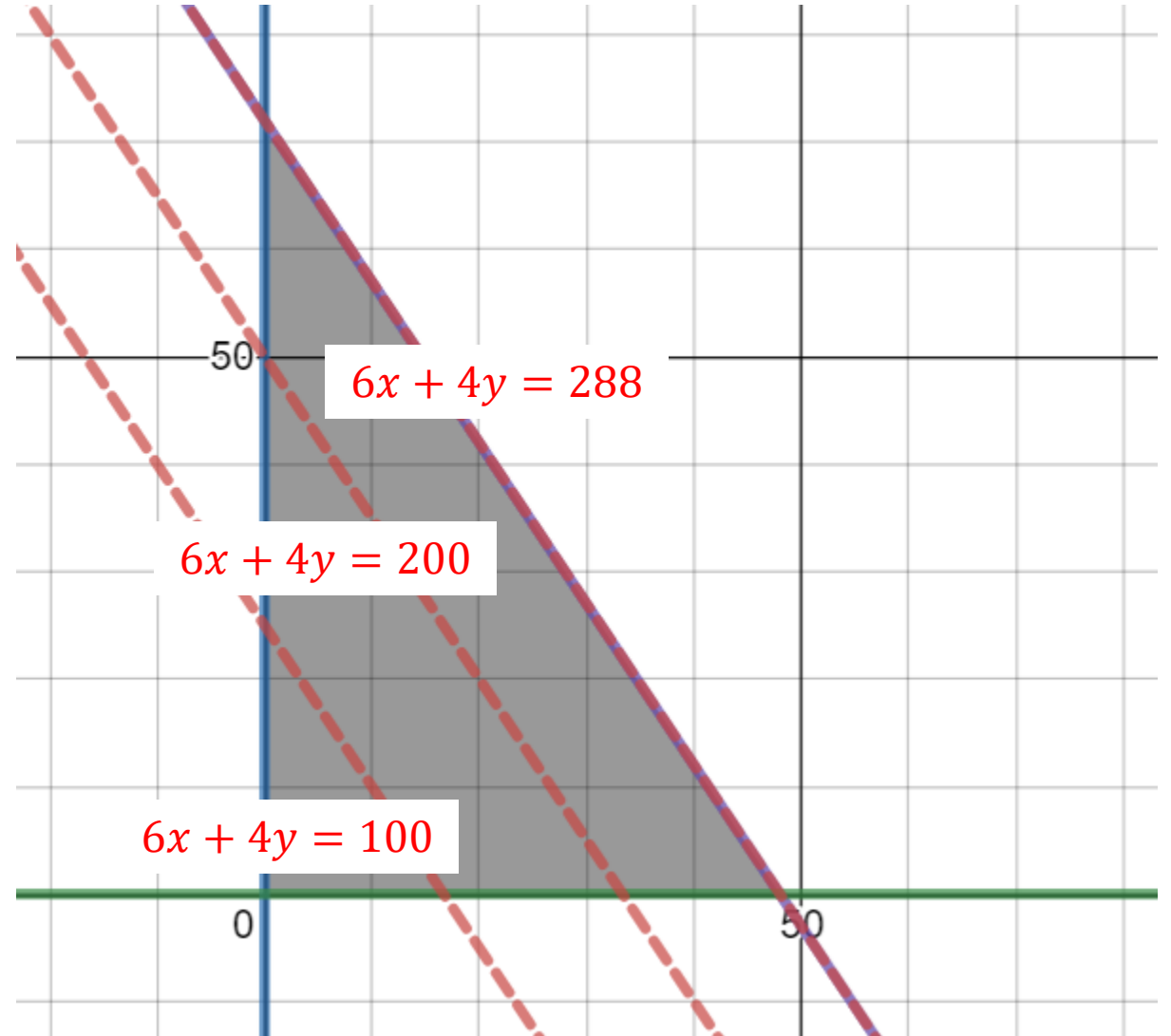
If our objective is parallel

maximise $P = 6x + 4y$
subject to:

$$\begin{aligned} 3x + 2y &\leq 144 \\ x &\geq 0 \\ y &\geq 0 \end{aligned}$$

Optimal solution anywhere on the line $3x + 2y = 144$. Thus $P = 288$

Note that it still has an optimal solution at a vertex!



Optimality

Optimal solution always contained at a boundary, however may lie on the boundary

Optimal solution always exists at a vertex, these are known as Basic Feasible Solutions.

A Basic Feasible Solution (BFS) is the intersection of a number of boundaries.

Graphical illustration of 3 variables

Intersection of different planes with a cube

e.g.

minimise $P = x + 3y + z$

subject to:

$$x \leq 1$$

$$y \leq 1$$

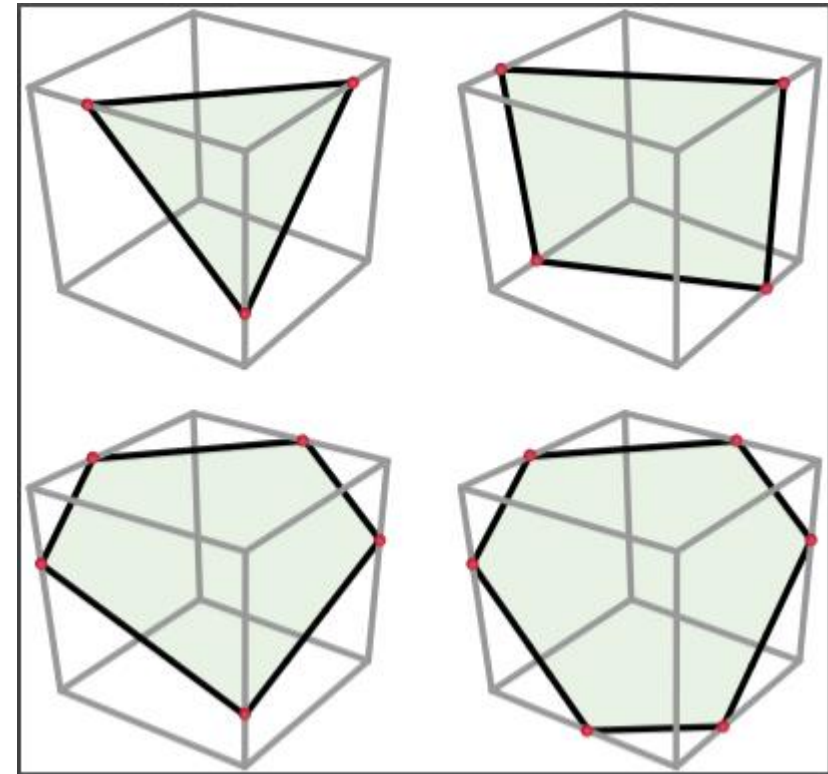
$$z \leq 1$$

$$x \geq 0$$

$$y \geq 0$$

$$z \geq 0$$

Each corner of the cube is a BFS.



Canonical Form (Matrices)

Our linear programming model

maximise $5x + 7y$

subject to:

$$3x + 2y \leq 144$$

$$x \geq 4y$$

$$x \geq 0$$

$$y \geq 0$$

maximise $5x + 7y$

subject to:

$$3x + 2y \leq 144$$

$$-x + 4y \leq 0$$

and

$$x \geq 0$$

$$y \geq 0$$

maximise $[5 \ 7] \begin{bmatrix} x \\ y \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Canonical Form (Matrices)

Let

$$\mathbf{c} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
subject to:
 $\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$
and
 $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Canonical Form

maximise $\mathbf{c}^T \mathbf{x}$
subject to: $\mathbf{Ax} \leq \mathbf{b}$
and $\mathbf{x} \geq \mathbf{0}$

In general

Let

$$\mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Canonical Form

maximise $\mathbf{c}^T \mathbf{x}$
subject to: $\mathbf{Ax} \leq \mathbf{b}$
and $\mathbf{x} \geq \mathbf{0}$

What are m and n for the example?

e.g

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \\ 100 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

maximum and minimum equivalence

maximise $\mathbf{c}^T \mathbf{x}$
subject to: $\mathbf{Ax} \leq \mathbf{b}$
and $\mathbf{x} \geq \mathbf{0}$

Is equivalent to...

minimise $-\mathbf{c}^T \mathbf{x}$
subject to: $\mathbf{Ax} \leq \mathbf{b}$
and $\mathbf{x} \geq \mathbf{0}$

Additional homework, why?

Solution via Python

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We are maximising.

The Python function `linprog` minimises!

Notice that in the Python code we set `c = [-5, -7]`

```
1 from scipy.optimize import linprog
2
3 # cost function (row vector) - to minimise we need the negatives
4 c = [-5, -7]
5
6 # linear inequalities
7 # (no. rows in A = no. rows in b)
8 A_ub = [[3, 2],
9         [-1, 4]]
10 b_ub = [144, 0]
11
12 # call the linprog function
13 results = linprog(c, A_ub, b_ub)
14
15 print(results.message)
16 # if results are a success then print out the optimal solution
17 if results.success:
18     for k in range(len(results.x)):
19         print(f"x[{k+1}] = {results.x[k]:.2f}")
20     print(f"fval = {results.fun:.2f}")
```

[scipy.optimize.linprog — SciPy v1.11.3 Manual](#)

Solution via Python

We will use a friendlier way to express this next week, however it important to:

- get to understand the canonical form
- think about this in terms of Linear Algebra

Tutorial

- Solve a linear program with Python
- Solve a linear program with Excel
- Model a simple LP problem and solve
- Model a more complicated LP, we will start the beginning of the next lecture with a discussion about this.